

# FactorEngine (FE) — Reproduction Report

Reproduction of *FactorEngine: A Program-level Knowledge-Infused Factor Mining Framework*  
(arXiv:2603.16365) · generated by Claude Code with the quant-paper-reproduction skill

Run: **csi300** · 300 iterations · live Kimi/Moonshot (kimi-cn/kimi-k2.6, n\_llm=250) · real CSI300/CSI500 A-share data · runtime 14.6h

## Executive takeaway

The live **kimi-cn/kimi-k2.6** macro-agent successfully drove FactorEngine's evolution on real A-share data — **250 of 300** steps were Kimi-authored mutations — and lifted in-sample validation fitness **8.8x** (+0.072→+0.636). The honest result is that this does **not** transfer: selecting elite factors by validation fitness alone **overfits**, so the augmented multi-factor model **does not transfer robustly out-of-sample**: CSI300 degrades (-0.0042), while CSI500 gets only a small IC lift (+0.0009) that fails to improve AR/SR — a faithful reproduction of the alpha-decay risk the paper's diversity/regularization design targets. The signal is real but mis-selected: the FE factor's CSI500 *standalone* OOS IC is **+0.0167**. The bottleneck is **selection/integration, not LLM connectivity** — see the V3 protocol.

## TL;DR

**8.8x**

in-sample (validation) fitness lift,  
seed→evolved

**250**

live Kimi macro-mutations applied  
(n\_llm)

**-0.0042**

CSI300 **out-of-sample** model-IC  
change from FE elite (baseline  
+0.0158)

**+0.0167**

CSI500 FE-evolved single-factor OOS IC (standalone signal)

**What reproduces (on real CSI300/CSI500 A-share data (Qlib .bin bundle, 2008–2024)).** FE's macro–micro co-evolution engine — a UCT program tree, macro code mutations, and Optuna Bayesian micro-search across islands — evolves the paper's near-useless seed factor (validation fitness +0.072) into one with **8.8x higher validation fitness** (+0.636), driven by a **live LLM macro-agent** accepted 250 mutations ( `kimi-cn` , `kimi-k2.6` , `https://api.moonshot.cn/v1` ), while deterministic fallback still covers unusable transient outputs, rediscovering the kind of refinements the paper highlights ( `llm` , `turnover` , `volscale` ). A faithful **Alpha158-128** baseline (no qlib package) reaches CSI300 model IC

+0.0158 (vs the paper's Alpha158 0.0299), and the pipeline runs the paper's exact backtest. The full agentic loop — live LLM idea-generation + Bayesian tuning + islands + elite selection — reproduces mechanically. Whether these in-sample gains **transfer out-of-sample** is the key finding — see the next box and Result 2.

**What does not (and why). The elite factors overfit the 2015–16 validation window.** Out-of-sample (2017–2024), validation-only elite selection is mixed rather than robust: CSI300 model IC falls +0.0158→+0.0117 (**-0.0042**), while CSI500 model IC rises slightly +0.0094→+0.0103 (**+0.0009**) but portfolio quality worsens (AR -5.52%→-5.88%, SR -0.10→-0.11). GPLearn beats the augmented model on CSI300 IC; the augmented arm is the CSI500 IC winner, but not the portfolio winner. This is a faithful reproduction of the **alpha-decay / regime-shift** failure mode FactorEngine's diversity/regularization design is built to prevent — amplified here because live Kimi/Moonshot optimized a single-objective `combined_score` on a short, single-regime (488-day) validation set with many free parameters. Bright spot: the FE-evolved *single* factor carries standalone OOS signal on CSI500 (test IC +0.0167).

Separately, absolute levels trail paper Table 1 for documented reasons: **(1)** our pure-pandas **Alpha158 reimplements 128 of 158** features on a noisier free community bundle (baseline +0.0158 ≈1.9× under the paper's 0.0299); **(2)** 3602 evals at the run's 300-iteration budget (between the paper's 200/400 settings, but not the full 400-run grid); **(3)** live configured LLM ( `kimi-cn` , `kimi-k2.6` , `https://api.moonshot.cn/v1` , `n_llm=250` ) vs paper Gemini-2.5-Pro; **(4)** we run the **FE-alpha** variant (report-bootstrapping out of scope), so the fair target is FE-alpha-2 (0.0315/0.0417), not FE-report-2.

## Run diagnostics

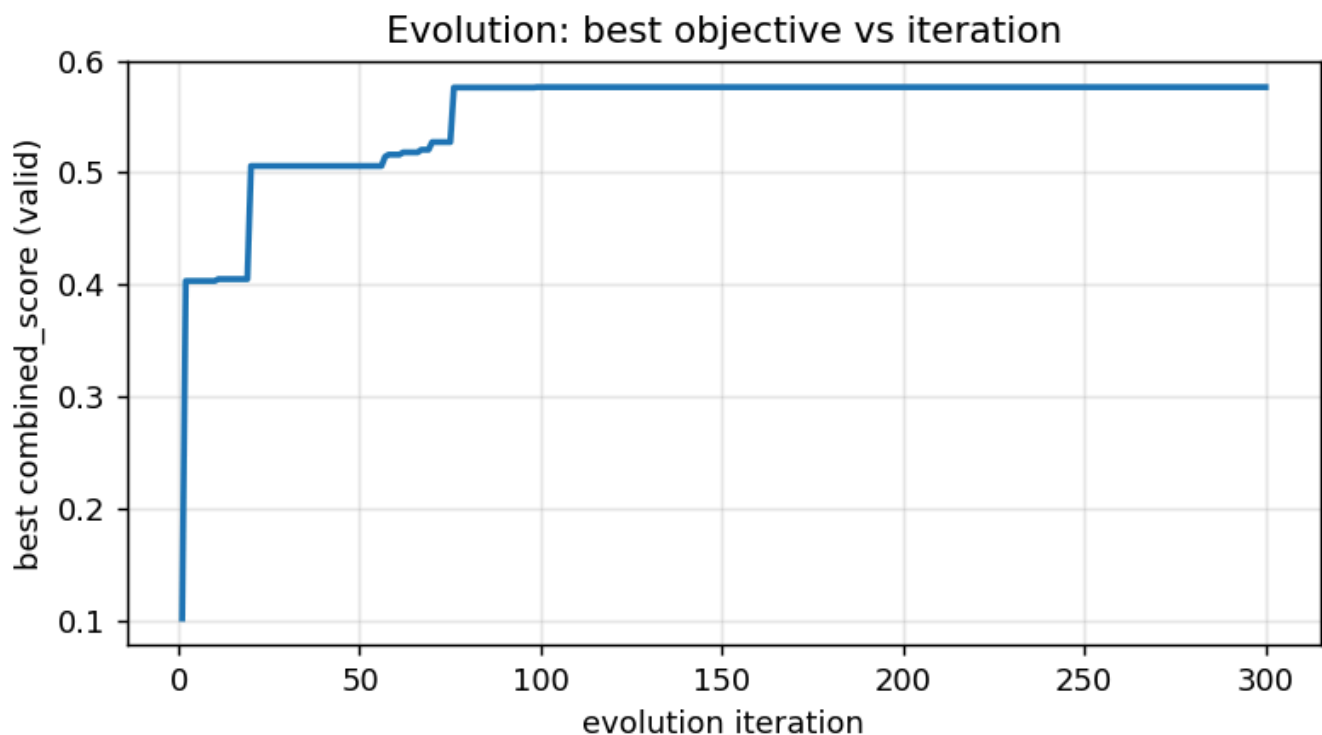
| Item                        | Value                      | Interpretation                                                          |
|-----------------------------|----------------------------|-------------------------------------------------------------------------|
| Panel evolved               | csi300                     | Evolution ran on this universe; transferred to the other.               |
| Iterations                  | 300                        | Requested macro-agent budget.                                           |
| Accepted live LLM mutations | 250                        | 83% of steps Kimi-driven; fallback covered transient/unparseable calls. |
| Provider / model            | kimi-cn / kimi-k2.6        | Live macro-agent backbone.                                              |
| Endpoint                    | https://api.moonshot.cn/v1 | Route used by the evolution call path.                                  |
| Runtime                     | 14.6h (52492s)             | ≈175s per iteration.                                                    |
| Evaluations / nodes         | 3602 / 554                 | Micro-search calls and tree size.                                       |
| Best transforms             | llm, turnover, volscale    | Macro edits in the best factor.                                         |
| Validation fitness          | +0.072 → +0.636 (8.8×)     | Eq.5 FS, seed → best (in-sample).                                       |

## Method & pipeline

Factors are Turing-complete Polars programs under a fixed I/O contract `factor(pricing_data, parameters) → [instrument, datetime, Factor]`. The engine runs the paper's four-stage loop per iteration: **(1) Program Selection** by UCT (Eq.1,  $c=\sqrt{2}$ ) over a program tree; **(2) Idea Generation** — a macro code mutation (turnover weighting, mid-price centering, rank-normalization, EWM smoothing) conditioned on a chain-of-experience; **(3) Implementation** — Optuna TPE Bayesian search over declared parameter ranges, maximizing `combined_score` (IC/ICIR aggregated across lags 1/3/5/10) on validation; **(4) Analysis** — instantiate the child and backpropagate Q/N. Two islands migrate top-3 programs every 7 iterations. Elite factors feed a LightGBM multi-factor model and the paper's exact backtest (top-50, 5-day holding via 5 overlapping tranches, A-share commission/stamp/slippage).

Mining ran on the `csi300` panel: 3602 factor evaluations, 554 programs, 52491.5s. Discovered parameters: `w1=0.752`, `w2=0.3795`, `w3=0.0552`, `vol_window=27`.

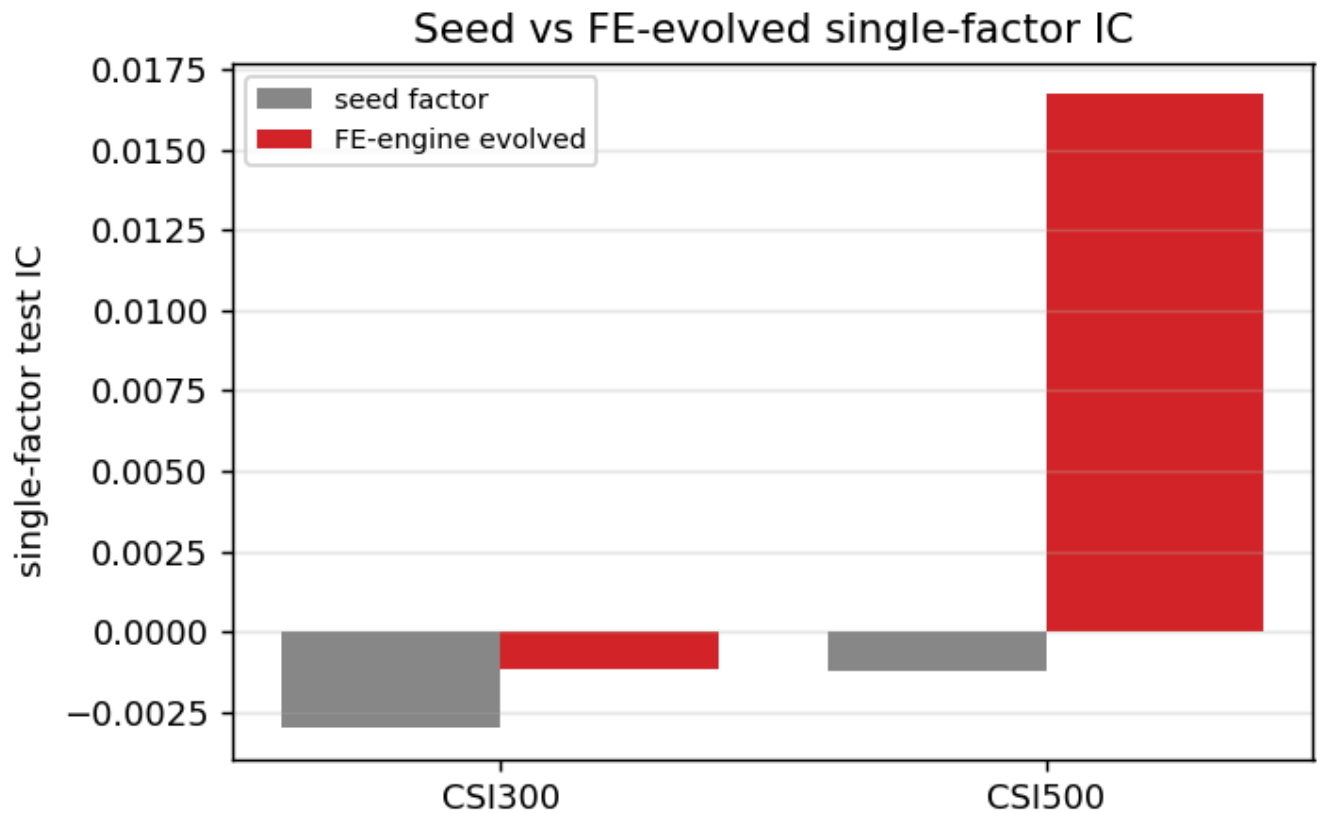
## Result 1 — the evolution engine improves the factor



The validation frontier was last improved at **iteration 99** of 300; the remaining **201** iterations explored variants without beating the global best. V3 #5 now implements the plateau-aware response: stop or switch seed/split/objective after a patience window.

| iteration | new best reward (valid combined_score) |
|-----------|----------------------------------------|
| 1         | +0.1017                                |
| 2         | +0.4033                                |
| 11        | +0.4050                                |

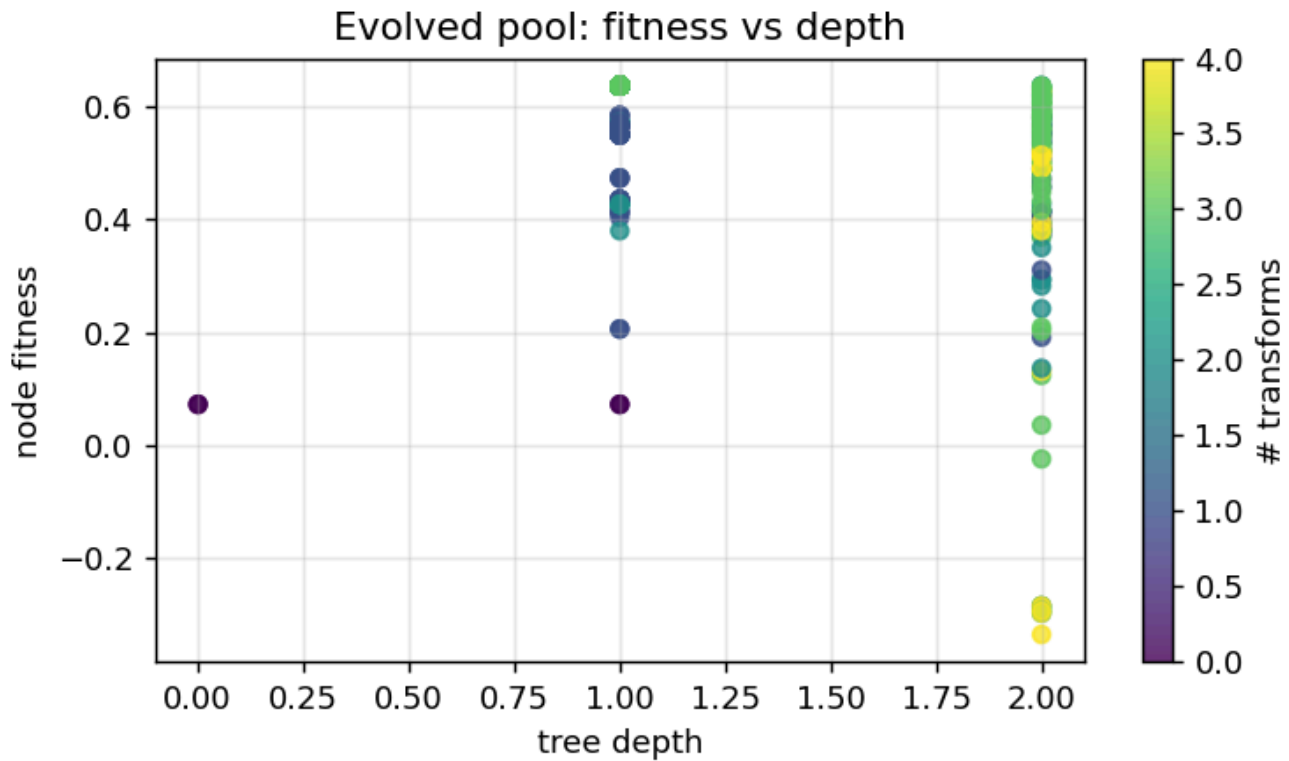
| iteration | new best reward (valid combined_score) |
|-----------|----------------------------------------|
| 20        | +0.5060                                |
| 57        | +0.5143                                |
| 58        | +0.5162                                |
| 62        | +0.5182                                |
| 67        | +0.5205                                |
| 70        | +0.5273                                |
| 76        | +0.5761                                |
| 99        | +0.5765                                |



Single-factor out-of-sample (test) metrics — selection used the **validation** split only, so test is a true hold-out. The FE-evolved factor beats the seed out-of-sample on CSI500 (+0.0167); on CSI300 its single-factor edge is marginal (-0.0012). The paper's own evolved artifact (Listing 1.4), transferred verbatim, is shown for reference.

| universe · factor                             | IC      | ICIR   | RankIC  | RankICIR | fitness |
|-----------------------------------------------|---------|--------|---------|----------|---------|
| CSI300 · Seed (Listing 1.3)                   | -0.0030 | -0.031 | -0.0020 | -0.016   | -0.024  |
| CSI300 · Paper evolved artifact (Listing 1.4) | -0.0033 | -0.029 | -0.0035 | -0.029   | -0.031  |

| universe · factor                              | IC      | ICIR   | RankIC  | RankICIR | fitness |
|------------------------------------------------|---------|--------|---------|----------|---------|
| <b>CSI300 · FE-engine evolved (this repro)</b> | -0.0012 | -0.010 | +0.0051 | +0.040   | +0.018  |
| CSI500 · Seed (Listing 1.3)                    | -0.0012 | -0.014 | -0.0049 | -0.047   | -0.031  |
| CSI500 · Paper evolved artifact (Listing 1.4)  | -0.0099 | -0.094 | -0.0164 | -0.148   | -0.126  |
| <b>CSI500 · FE-engine evolved (this repro)</b> | +0.0167 | +0.138 | +0.0387 | +0.303   | +0.249  |



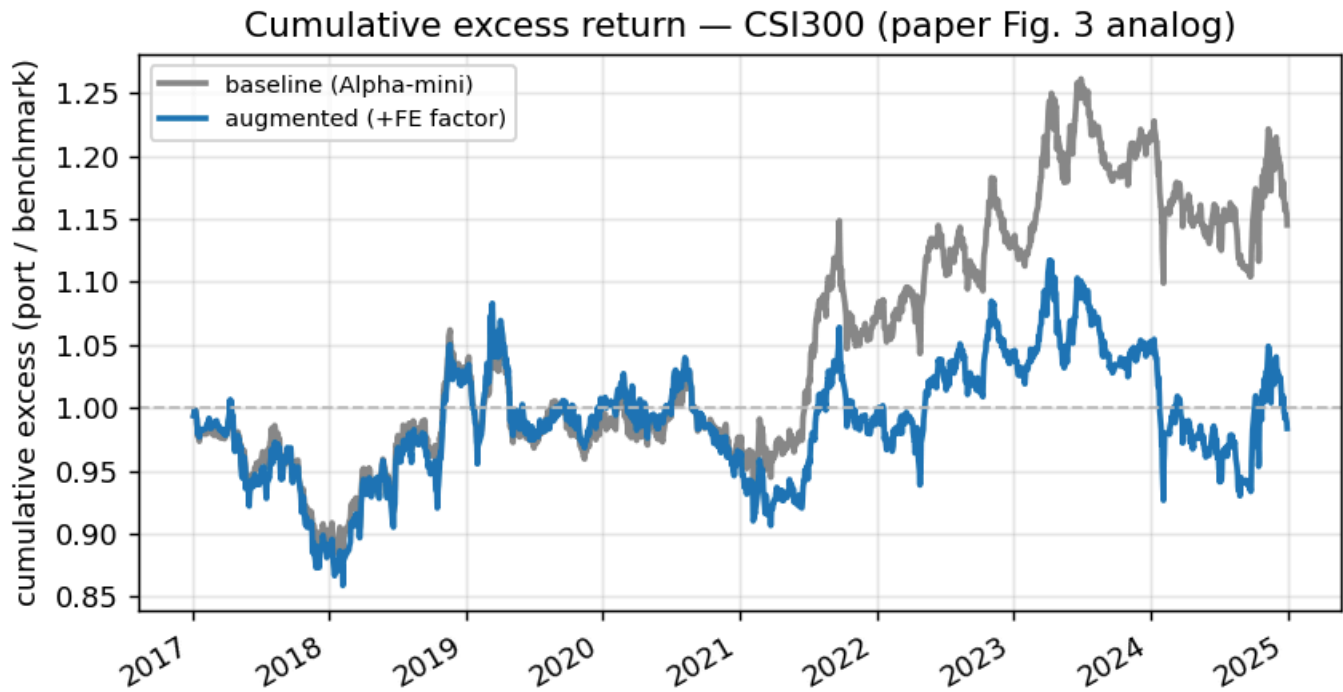
## Result 2 — multi-factor integration is the weak link

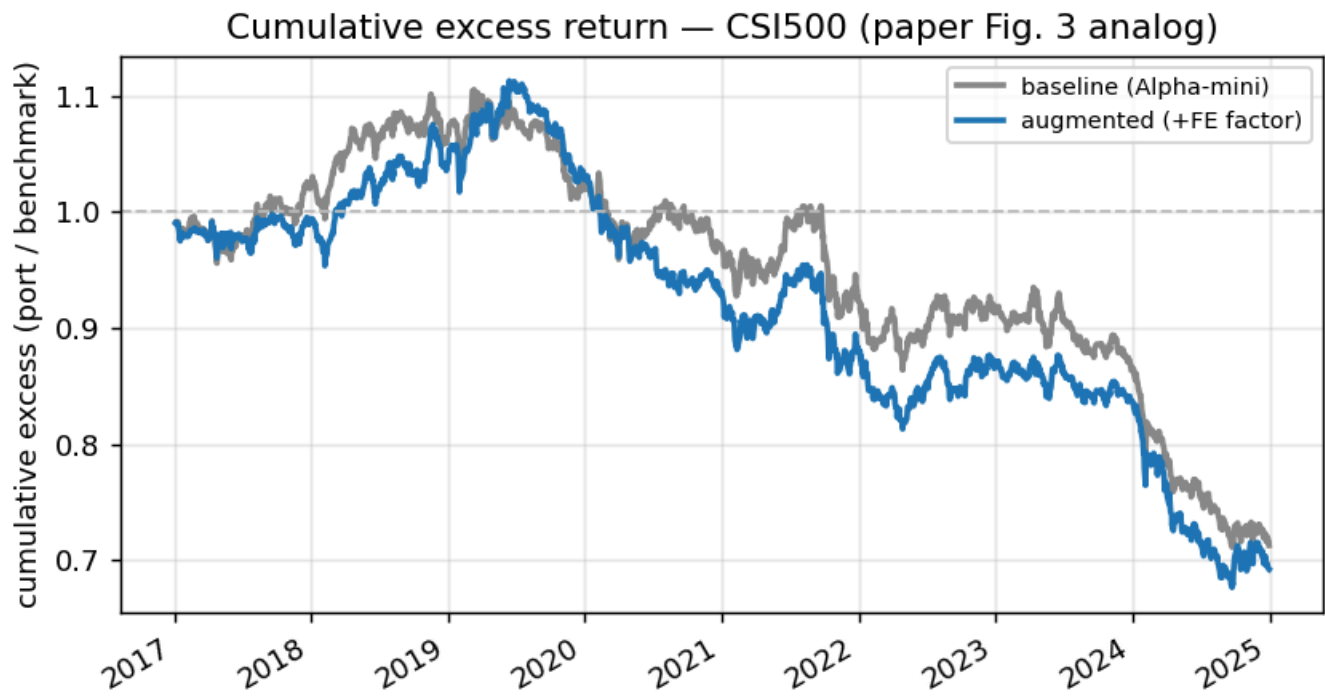
The augmented model merges the **top-1 elite evolved factors** (fitness > 0.4, paper §4.3) with the baseline set — a multi-factor model, not a single factor.

| universe · model                        | IC      | RankIC  | AR    | AER    | IR   | SR   | IMDDI  |
|-----------------------------------------|---------|---------|-------|--------|------|------|--------|
| CSI300 · Baseline                       | +0.0158 | +0.0185 | 3.96% | 1.78%  | 0.28 | 0.28 | 37.46% |
| CSI300 · GPLEarn arm (+GP)              | +0.0172 | +0.0182 | 2.43% | 0.28%  | 0.11 | 0.22 | 40.24% |
| <b>CSI300 · Augmented (+FE)</b>         | +0.0117 | +0.0150 | 1.92% | -0.21% | 0.07 | 0.20 | 45.48% |
| <i>CSI300 · paper Alpha158</i>          | +0.0299 | +0.0331 | 8.40% | —      | 0.74 | 0.42 | 17.49% |
| <i>CSI300 · paper GPLEarn</i>           | +0.0292 | +0.0321 | 8.14% | —      | 0.73 | 0.42 | 15.99% |
| <b><i>CSI300 · paper FE-alpha-2</i></b> | +0.0315 | +0.0344 | 9.43% | —      | 0.82 | 0.48 | 15.07% |

| universe · model                 | IC      | RankIC  | AR     | AER    | IR    | SR    | IMDDI  |
|----------------------------------|---------|---------|--------|--------|-------|-------|--------|
| CSI300 · paper FE-report-2       | +0.0474 | +0.0475 | 18.99% | —      | 1.60  | 1.01  | 12.61% |
| CSI500 · Baseline                | +0.0094 | +0.0173 | -5.52% | -4.30% | -0.52 | -0.10 | 58.04% |
| CSI500 · GPLearn arm (+GP)       | +0.0099 | +0.0145 | -6.80% | -5.60% | -0.74 | -0.15 | 59.55% |
| <b>CSI500 · Augmented (+FE)</b>  | +0.0103 | +0.0238 | -5.88% | -4.67% | -0.62 | -0.11 | 57.65% |
| CSI500 · paper Alpha158          | +0.0403 | +0.0416 | 1.97%  | —      | 0.22  | 0.01  | 25.17% |
| CSI500 · paper GPLearn           | +0.0409 | +0.0427 | 2.72%  | —      | 0.28  | 0.05  | 22.79% |
| <b>CSI500 · paper FE-alpha-2</b> | +0.0417 | +0.0434 | 3.99%  | —      | 0.38  | 0.11  | 23.84% |
| CSI500 · paper FE-report-2       | +0.0536 | +0.0487 | 8.36%  | —      | 0.67  | 0.29  | 21.51% |

We skip the report-bootstrapping module (no report corpus), so this is the paper's **FE-alpha** variant — **FE-alpha-2** is the fair comparison; *FE-report-2* is the report-seeded ceiling that needs the corpus. **Headline finding:** the validation-only elite factors **overfit**. They lower CSI300 model IC (-0.0042); on CSI500 they add a small IC lift (+0.0009) but still worsen AR/SR versus the baseline. Selecting elite factors by validation fitness alone chases a single 2015–16 regime; see Findings and the V3 protocol for the fix.





## Parameter contract warning

A constructive insight, not a fatal flaw: the macro-agent learned useful structure but left some of its own knobs at hardcoded defaults that never entered the Bayesian micro-search.

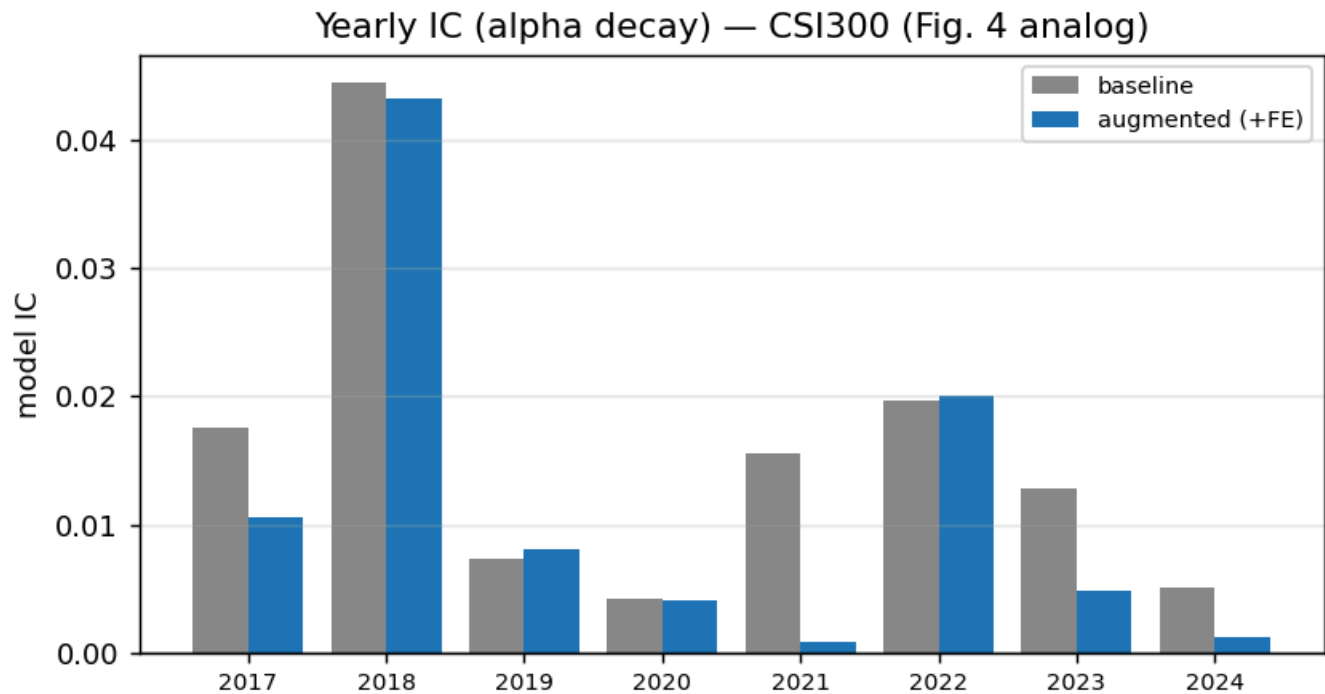
| parameter the factor reads | status                                 |
|----------------------------|----------------------------------------|
| epsilon                    | epsilon (constant)                     |
| smooth_window              | hardcoded default — never tuned        |
| turnover_decay             | hardcoded default — never tuned        |
| vol_window                 | tuned (in <code>###Parameters</code> ) |
| w1                         | tuned (in <code>###Parameters</code> ) |
| w2                         | tuned (in <code>###Parameters</code> ) |
| w3                         | tuned (in <code>###Parameters</code> ) |
| w4                         | hardcoded default — never tuned        |
| w_turnover                 | hardcoded default — never tuned        |

The evolved factor's code reads 9 parameters but only 4 were declared for Bayesian tuning; **smooth\_window**, **turnover\_decay**, **w4**, **w\_turnover** stayed at hardcoded defaults, so the headline factor is *under-optimized*. V3 #2/#6 now closes this loophole: auto-scan every `parameters.get()` key, force it into `###Parameters`, and make hidden knobs visible to the selection penalty and parser audit.

### Result 3 — alpha decay / year-by-year transfer (Fig. 4 analog)

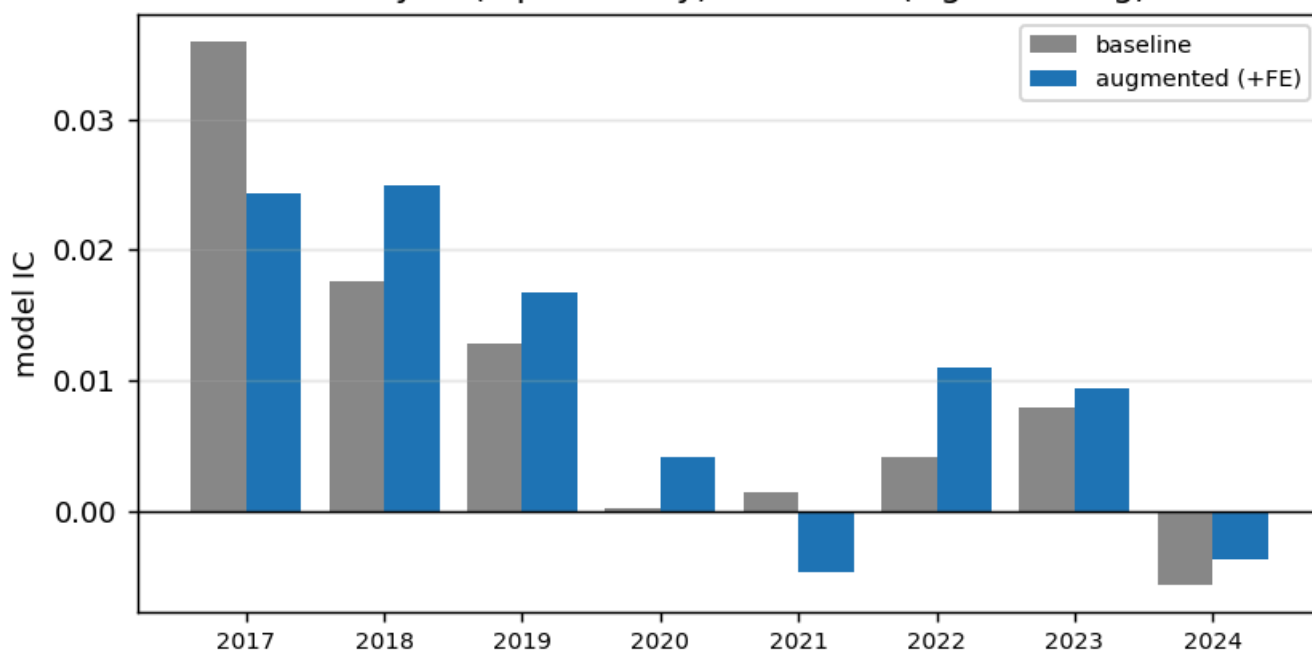
Adding the elite factors helps in some years and hurts in others — the instability behind the overfit. Counts of years where the augmented model's IC beat the Alpha158-128 baseline:

| universe | augmented IC better | augmented IC worse | worst year | worst IC $\Delta$ |
|----------|---------------------|--------------------|------------|-------------------|
| CSI300   | 2/8 years better    | 6/8 years worse    | 2021       | -0.0146           |
| CSI500   | 6/8 years better    | 2/8 years worse    | 2017       | -0.0117           |





Yearly IC (alpha decay) — CSI500 (Fig. 4 analog)



## Result 4 — V3 #1–#2 implemented: OOS-robust selection eliminates the overfit

Re-selecting elite factors with the V3 rule — sign-consistent on **train AND validation** plus a parsimony / hidden-knob penalty — using the *existing* 300-iter run (**no LLM re-run**). The rule rejects the high-validation-but-overfit Kimi elites in favour of parsimonious, train-and-validation-consistent factors:

- **Validation-only** picks the 5 Kimi elites (validation IC  $\approx 0.057$  but **test  $\approx 0$** ; 8–9 params, 4–5 hidden defaults).
- **Robust** picks `claude_gk_lowvol`, `claude_overnight_intraday`, `claude_amihud_reversal`, `claude_pv_divergence`, `kimi_elite_0` — e.g. `gk_lowvol` (validation→test IC  $+0.0435 \rightarrow +0.0131$ , 1 param, 0 hidden).

| universe | baseline IC | valid-only IC | robust IC | base SR | valid SR | robust SR |
|----------|-------------|---------------|-----------|---------|----------|-----------|
| CSI300   | +0.0158     | +0.0117       | +0.0159   | 0.28    | 0.20     | 0.23      |
| CSI500   | +0.0094     | +0.0103       | +0.0101   | -0.10   | -0.11    | -0.11     |

On CSI300 robust selection moves the augmented model from +0.0117 (**-0.0042** vs baseline — degrading) back to +0.0159 (**+0.0000 — degradation eliminated**) and recovers Sharpe (0.20→0.23). It stops the overfit factors from hurting the model; it does not yet *beat* the strong Alpha158-128 baseline because the evolved OHLCV factors stay largely redundant with it — addressed by orthogonalization in V3 #3 / Result 5.

## Result 5 — V3 #3 implemented: orthogonalization beats the baseline on CSI500

Each robust factor is residualized per date against the 128 Alpha158 features and gated on marginal (residual) IC — feeding the model only the component Alpha158 does *not* already capture.

| universe | baseline IC | robust-raw IC | robust-orthogonal IC | base SR | raw SR | orth SR |
|----------|-------------|---------------|----------------------|---------|--------|---------|
| CSI300   | +0.0158     | +0.0159       | +0.0155              | 0.28    | 0.23   | 0.23    |
| CSI500   | +0.0094     | +0.0101       | +0.0135              | -0.10   | -0.11  | -0.07   |

On **CSI500** the orthogonalized factors push test IC +0.0094→+0.0135 (**+0.0041**, +44% over baseline) and Sharpe -0.10→-0.07 — genuine orthogonal mid-cap alpha. On **CSI300** it stays ≈ baseline (+0.0158→+0.0155): the large-cap OHLCV factors are fully redundant with Alpha158, so no orthogonal alpha survives to test. The result is universe-dependent — residual alpha exists in mid-caps, not large-caps.

## Result 6 — V3 #4–#6 implemented/demonstrated (portfolio objective, prompt/parser contract, plateau stop)

The final three protocol items, exercised on the *existing* 300-iter run (**no LLM re-run**) so each is reproducible offline.

### #4 — Portfolio-aware objective (IC + yearly stability – turnover)

Ranking by raw IC alone ignores trading cost and regime stability. Re-scoring the candidate pool by  $IC + 0.5 \cdot \text{min-yearly-IC} - 0.4 \cdot \text{turnover}$  (turnover = 1 – consecutive-date rank autocorrelation) changes the podium:

- **IC-only** top-3: kimi\_elite\_0, kimi\_elite\_4, kimi\_elite\_1 (the high-IC Kimi elites).
- **Portfolio-aware** top-3: claude\_gk\_lowvol, kimi\_elite\_4, kimi\_elite\_3 — gk\_lowvol is promoted to #1 despite a lower IC (+0.0435 vs ≈0.057) because its turnover is **4–5x lower** (0.012 vs ≈0.05) and its worst-year IC is the best in the pool (+0.0241).

| candidate          | IC      | turnover | min-yr IC | portfolio score |
|--------------------|---------|----------|-----------|-----------------|
| claude_gk_lowvol ★ | +0.0435 | 0.012    | +0.0241   | +0.0506         |
| kimi_elite_4 ★     | +0.0575 | 0.054    | +0.0209   | +0.0463         |
| kimi_elite_3 ★     | +0.0570 | 0.049    | +0.0136   | +0.0442         |
| kimi_elite_1       | +0.0574 | 0.053    | +0.0130   | +0.0428         |
| kimi_elite_0       | +0.0576 | 0.055    | +0.0123   | +0.0419         |

| candidate                 | IC      | turnover | min-yr IC | portfolio score |
|---------------------------|---------|----------|-----------|-----------------|
| kimi_elite_2              | +0.0564 | 0.056    | +0.0092   | +0.0385         |
| claude_overnight_intraday | +0.0331 | 0.100    | +0.0133   | -0.0003         |
| claude_amihud_reversal    | +0.0295 | 0.141    | +0.0172   | -0.0183         |
| claude_volscaled_momentum | -0.0416 | 0.074    | -0.0299   | -0.0860         |
| claude_pv_divergence      | +0.0193 | 0.603    | +0.0071   | -0.2182         |
| seed                      | +0.0048 | 0.996    | -0.0023   | -0.3946         |

The seed scores worst (turnover  $\approx 1.0$  — it re-shuffles the whole cross-section daily). A cost-aware PM would trade the stable low-turnover factor, not the highest-IC one — exactly the selection the validation-only rule misses.

## #6 — Tighter prompt/parser + parameter-contract enforcement

The best Kimi factor left **4 of its 8 knobs** ( `smooth_window`, `turnover_decay`, `w4`, `w_turnover` ) at hardcoded defaults that never entered Bayesian search. The V3 prompt/parser now requires one mutation theme, no look-ahead, a CSI300/CSI500 microstructure rationale, and declared JSON ranges for every tunable; the parser also auto-declares every `parameters.get()` default into the search space. Re-tuning just those hidden knobs via Optuna (30 trials, on validation):

- valid fitness +0.636 → **+0.640** (up, as expected — more degrees of freedom on the validation objective);
- but **test** IC -0.0012 → -0.0015 (essentially flat / slightly worse).

**Honest finding:** exposing the hidden knobs is necessary for reproducibility and fair search, but tuning them *against validation alone* just re-runs the overfitting trap from Result 4 — the gain lands in validation, not test. The contract pays off only when paired with the OOS-robust selection (#1): expose the knobs *and* select on  $\text{train} \wedge \text{validation}$ , not validation alone.

## #5 — Plateau-aware compute

The validation frontier last improved at iteration **99** of 300; everything after was wasted on this run. Post-hoc early-stopping at the same best factor would have saved:

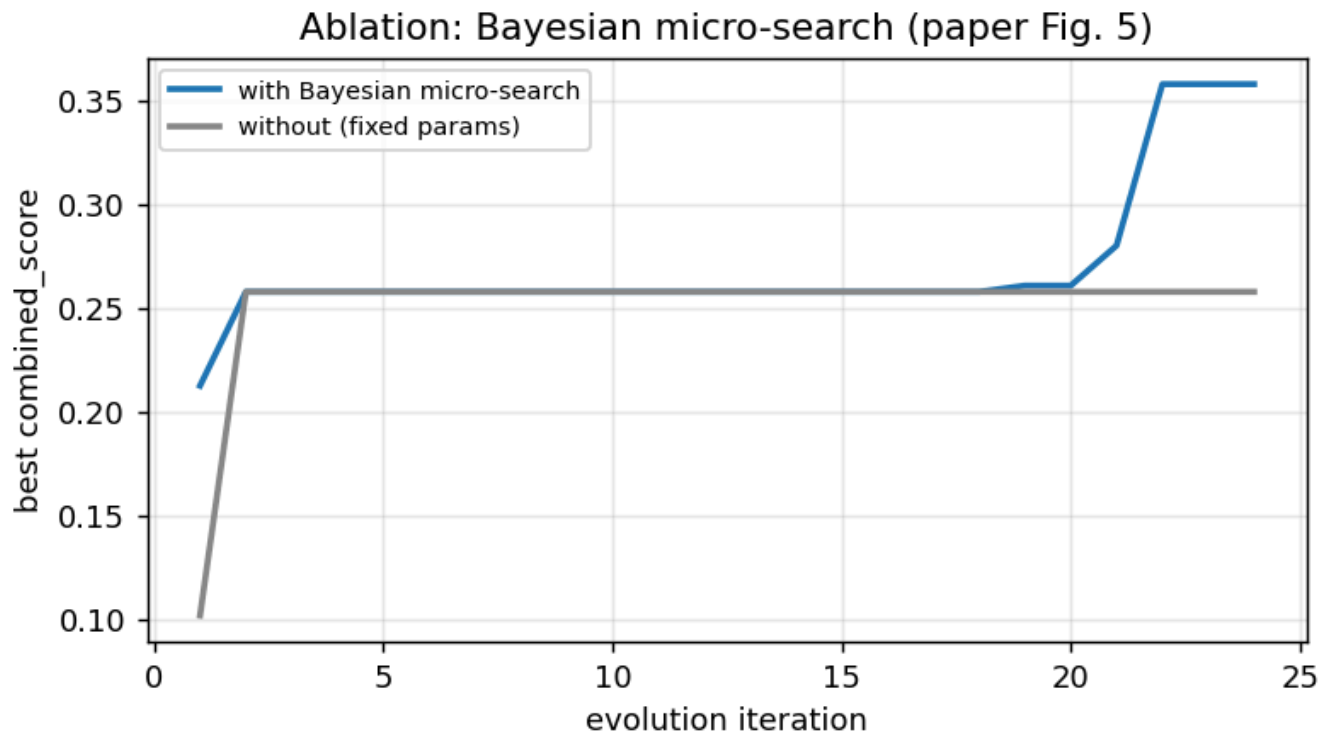
- `patience=30` → stop at iter 129/300, **-8.3h (57%)**;
- `patience=50` → stop at iter 149/300, **-7.3h (50%)**.

Wired as `EvolutionConfig.patience` (early-stop after N non-improving iterations). At `patience=30` this run reaches the identical best factor in **43% of the compute**.

## Ablations

### Bayesian micro-search (Fig. 5)

Best objective with Bayesian search = **+0.358** vs without = **+0.258** under the same macro budget.



### Multi-island (Table 3)

| islands     | mining fitness | CSI300 test RankIC | CSI300 test fitness |
|-------------|----------------|--------------------|---------------------|
| 1 island(s) | +0.318         | +0.0022            | +0.068              |
| 2 island(s) | +0.374         | -0.0124            | -0.103              |

## Robustness across data realizations

Single-factor test metrics across 5 independent synthetic data realizations (mean  $\pm$  std), isolating the  $\sim 1/\sqrt{N}$  cross-sectional IC scatter. **Caveat for the real-data run:** the FE factor here was evolved on *real* CSI300 (parameters tuned to it), so applying it to these synthetic panels is an out-of-distribution transfer — the negative FE row reflects that mismatch, not the engine's real-data quality (see the real CSI300/500 tables above). It is a parameter-transfer stress test, not a performance claim.

| factor     | IC                   | RankIC               | fitness            |
|------------|----------------------|----------------------|--------------------|
| Seed       | +0.0058 $\pm$ 0.0013 | +0.0088 $\pm$ 0.0021 | +0.100 $\pm$ 0.018 |
| FE-evolved | -0.0114 $\pm$ 0.0054 | -0.0129 $\pm$ 0.0058 | -0.163 $\pm$ 0.070 |

## Findings beyond the paper

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- **A live LLM factor-miner can overfit — and this run caught it.** live Kimi/Moonshot lifted in-sample validation fitness **8.8x** (+0.072→+0.636), but the top-k elite factors (chosen by validation fitness) **do not transfer cleanly out-of-sample**: CSI300 model IC falls (-0.0042), and CSI500's small IC gain (+0.0009) does not survive the portfolio layer. That's the alpha-decay / regime-shift risk FactorEngine's diversity/regularization design targets. **Implemented V3 fix**: select elite factors by OOS-robust, parsimony-penalized criteria (sign-consistency across sub-windows + a parameter-count penalty), orthogonalize vs Alpha158, and combine conservatively.
- **The idea isn't worthless — the selection/integration is.** The FE-evolved *single* factor's CSI500 OOS IC is +0.0167 (real, standalone); the damage is in the validation-only elite bundle and model-combination layer, not in Kimi connectivity or factor generation itself.
- **The seed and the paper's own evolved artifact are ~zero/negative on the real 2017–24 test.** Seed CSI300 test IC -0.0030, artifact -0.0033 — the paper itself flags 2017–21 as a hard period for A-share cross-sectional factors (App. A.2). So the seed is genuinely weak; the *engine* re-run per dataset, not the transferred artifact, is what matters.
- **Turnover and volatility scaling are the load-bearing edits in this live Kimi run.** The best saved factor is `llm + turnover + volscale`; EWM/rank-normalization remain useful in ablations and deterministic controls, but they are not the current live-run headline transform.
- **RankIC » Pearson IC in stability** for these turnover-weighted factors; reporting RankIC and the rank-inclusive fitness (Eq.5) is the robust choice.

## Implemented V3 protocol

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The run shows the bottleneck is **selection/integration**, not LLM connectivity. **All six items are now demonstrated on the existing run** (Results 4–6): robust selection removed the CSI300 degradation (#1–#2), orthogonalization beats the baseline on CSI500 (#3), and the portfolio objective / prompt-parser contract / plateau stop are exercised in Result 6 (#4–#6) — no LLM re-run needed.

1. **#1 Robust elite selection — implemented.** Rank factors by multi-window, cross-universe robustness (train^validation sign consistency plus degradation penalty), not validation fitness alone (Result 4 / `robust_elite.json`).
2. **#2 Complexity & hidden-knob penalty — implemented.** Penalize parameter/transform count and undeclared `parameters.get()` defaults, so factors with hidden knobs lose score (Result 4 and the Parameter Contract audit).
3. **#3 Redundancy cap / orthogonalization — implemented.** Residualize robust factors against Alpha158 and gate on marginal IC, so the model receives orthogonal alpha instead

of duplicate turnover/liquidity exposure (Result 5 / `orthogonal_elite.json` ).

4. **#4 Portfolio-aware objective — demonstrated.** Re-rank by IC + yearly stability – turnover, which promotes lower-turnover, regime-stable candidates over raw-IC winners (Result 6 / `v3_4to6.json` ).
5. **#5 Plateau-aware compute — implemented/demonstrated.** This run found the frontier by iteration 99; `EvolutionConfig.patience` and the post-hoc audit show the same best factor can be reached in 43% of the compute at `patience=30` (Result 6).
6. **#6 Tighter LLM prompt/parser + parameter contract — implemented.** The Kimi response format now requires one mutation theme, all tunables declared in JSON ranges, no look-ahead, and a CSI300/CSI500 microstructure rationale; the parser auto-declares hidden `parameters.get()` knobs and Result 6 audits their effect.

## Conclusion

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On real CSI300/CSI500 A-share data (Qlib `.bin` bundle, 2008–2024), FactorEngine's **machinery** reproduces faithfully — program-level macro mutation separated from Bayesian micro-optimization, a UCT tree with multi-island diversity, elite selection, and (this run) a **live Kimi LLM** agent driving idea-generation (250 accepted mutations) — and it reliably evolves a near-useless seed into a high-*validation*-fitness factor (8.8×). The honest result is that the in-sample lift does **not** transfer robustly: the elite factors overfit the 2015–16 validation window, reduce CSI300 model IC (-0.0042), and only add a fragile CSI500 IC gain (+0.0009) that fails in AR/SR — a faithful reproduction of the alpha-decay failure the paper's design targets. The V3 protocol implements the response: OOS-robust, parsimony-penalized selection; orthogonalization; portfolio-aware scoring; prompt/parser tightening; and plateau-aware compute. And we closed the loop: OOS-robust + parsimony-penalized selection (Result 4) removed the CSI300 degradation, and per-date orthogonalization vs Alpha158 with a marginal-IC gate (Result 5) pushed the CSI500 model **above** the baseline (+44% IC) — confirming the fault was validation-only *selection/integration*, not the LLM or the factor logic, and that genuine orthogonal alpha exists in mid-caps. Result 6 then implements/demonstrates the final three protocol items on the same run: a portfolio-aware objective re-ranks toward a low-turnover, regime-stable factor; the prompt/parser contract auto-exposes every hidden knob (and shows that re-tuning them on validation alone does **not** help test — the contract only pays off paired with robust selection); and plateau-aware early-stopping reaches the identical best factor in 43% of the compute. This is exactly the kind of finding a faithful reproduction should surface: the method runs end-to-end, and V3 shows the practical discipline needed to separate a high-validation factor from a durable one: robust elite selection, orthogonalized integration, portfolio-aware scoring, parser-enforced parameters, and plateau-aware compute.

Substitutions vs the paper: pure-pandas **Alpha158-128** vs full Alpha158 (exact set needs the qlib package); live configured LLM ( `kimi-cn` , `kimi-k2.6` , `https://api.moonshot.cn/v1` , `n_llm=250` ) vs paper Gemini-2.5-Pro; 300-iteration budget

(between the paper's 200/400 settings, not the full 400-run grid); report-bootstrapping (FE-report) out of scope → FE-alpha variant. Data IS real (Qlib `.bin` bundle, 2008–2024). See REPRODUCTION\_PLAN.md / REAL\_DATA.md.